Particles 1

March 2026

1 Introduction

1.1 Special relativity

Given two inertial frames $S(t, x, y, z)$ and $S^{'}(t^{'}, x^{'}, y^{'}, z^{'})$, the equations that connect the spatial and temporal coordinates in the two frames are called transformations. These are the Lorentz transformations.
To be inertial, the two reference frames should be in rectilinear uniform relative motion.
We take the corresponding axes of the two frames in the same directions, and consider, as usual, the case in which the relative motion is in the x direction. At a certain moment, which we take as $t^{'} = t = 0$, the origins and the axis coincide. The frame $S^{'}$ moves relative to S with speed $\vec{v}$.
We introduce the following two dimensionless quantities relative to the motion in S of the origin $S^{'}$

$$\beta = \frac{\vec{v}}{c} \tag{1}$$

and

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}, \tag{2}$$

called the 'Lorentz factor'.
An event is defined by the four-vector of the coordinates $(ct, \vec{r})$. Its components in the two frames (t, x, y, z) and $(t^{'}, x^{'}, y^{'}, z^{'})$ are linked by the Lorentz transformations, in the considered particular case of relative motion in the x direction, as

$$\begin{cases} x^{'} = \gamma(x - \beta ct) \\ y^{'} = y \\ z^{'} = z \\ ct^{'} = \gamma(ct - \beta x). \end{cases} \tag{3}$$

The set of all Lorentz transformations, when joined to the rotations of the axes, form a group called the proper Lorentz group. The group contains the parameter c, a constant with the dimensions of the velocity. A physical entity moving at speed c in a reference frame moves with the same speed in any other one. In

other words, c in invariant under Lorentz transformations. It is the propagation speed of light and gravitational waves.
Of special importance is the energy-momentum vector $(E/c, \vec{p})$ of a free particle

$$\begin{cases} p_x^{'} = \gamma\left(p_x - \beta\frac{E}{c}\right) \\ p_y^{'} = p_y \\ p_z^{'} = p_z \\ \frac{E^{'}}{c} = \gamma\left(\frac{E}{c} - \beta p_x\right). \end{cases} \tag{4}$$

The transformations that give the component in S as functions of those in $S^{'}$ can be most simply obtained by changing the sign of the speed $\vec{v}$.
The norm of the energy-momentum is, as for the four-vectors, an invariant; it is the square of the mass of the system multiplied by the invariant factor c^4

$$m^2c^4 = E^2 - p^2c^2. \tag{5}$$

The most general relationship between the linear momentum $\vec{p}$, the energy E and the speed $\vec{v}$ is

$$\vec{p} = \frac{E}{c^2}\vec{v}, \tag{6}$$

which is valid both for bodies with zero and non-zero mass.
For massless particles, $m^2c^4 = E^2 - p^2c^2$ can be written as

$$pc = E. \tag{7}$$

The photon mass is exactly zero. Neutrinos have non-zero but extremely small masses in comparison with the other particles.
If $m \neq 0$, the energy can be written as

$$E = m\gamma c^2, \tag{8}$$

and $\vec{p} = \frac{E}{c^2}\vec{v}$ takes the equivalent form

$$\vec{p} = m\gamma\vec{v}. \tag{9}$$

1.2 The Law of Motion of a Particle

$$\vec{F} = \frac{d\vec{p}}{dt} = m\gamma\vec{a} + m\frac{d\gamma}{dt}\vec{v}, \tag{10}$$

taking the derivative, we obtain

$$m\frac{d\gamma}{dt}\vec{v} = m\frac{d\left(1 - \frac{v^2}{c^2}\right)^{-1/2}}{dt}\vec{v} = -m\frac{1}{2}\left(1 - \frac{v^2}{c^2}\right)^{-3/2}\left(-2\frac{v}{c^2}a_t\right)\vec{v} = m\gamma^3(\vec{a}\cdot\vec{\beta})\vec{\beta}.$$

Hence,

$$\vec{F} = m\gamma\vec{a} + m\gamma^3(\vec{a}\cdot\vec{\beta})\vec{\beta}. \tag{11}$$

To solve for the acceleration, we take the scalar product of the two members of the previous equation with β. We obtain

$$\vec{F} \cdot \vec{\beta} = m\gamma\vec{a} \cdot \vec{\beta} + m\gamma^3\beta^2\vec{a} \cdot \vec{\beta} = m\gamma(1 + \gamma^2\beta^2)\vec{a} \cdot \vec{\beta} = m\gamma^3\vec{a} \cdot \vec{\beta}.$$

Hence,

$$\vec{a} \cdot \vec{\beta} = \frac{\vec{F} \cdot \vec{\beta}}{m\gamma^3}$$

and, by substitution into $\vec{F} = m\gamma\vec{a} + m\gamma^3(\vec{a} \cdot \vec{\beta})\vec{\beta}$,

$$\vec{F} - (\vec{F} \cdot \vec{\beta})\vec{\beta} = m\gamma\vec{a}. \tag{12}$$

1.3 The Mass of a System of Particles: Kinematic Invariants

The mass of a system of particles is often called "invariant mass", but the adjective is useless; the mass is always invariant. The expression is simple only if the particles of the system do not interact amongst themselves. In this case, for n particles of energies E_i and momenta $\vec{p}_i$, the mass is

$$m = \sqrt{E^2 - P^2} = \sqrt{\left(\sum_{i=1}^{n} E_i\right)^2 - \left(\sum_{i=1}^{n} \vec{p}_i\right)^2}. \tag{13}$$

Consider the square of the mass, which we shall indicate by s, obviously an invariant quantity

$$s = E^2 - P^2 = \left(\sum_{i=1}^{n} E_i\right)^2 - \left(\sum_{i=1}^{n} \vec{p}_i\right)^2. \tag{14}$$

Notice that s cannot be negative

$$s \geq 0. \tag{15}$$

In the "center of mass" (CM), which is defined as the reference in which the total momentum is zero, we have that

$$s = \left(\sum_{i=1}^{n} E_i^*\right)^2, \tag{16}$$

where E^* are the energies in the CM. In other words, the mass of a system of non-interacting particles is also the energy in the CM frame.

1.3.1 Collisions

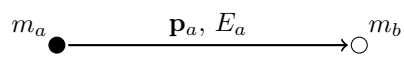
In the initial state, two particles, a and b, are present; in the final state we may have two particles (not necessarily a and b) or more. Call these c, d, e, $\cdots$. The process is

$$a + b \to c + d + e + \cdots . \tag{17}$$

If the final state contains only the initial particles, then the collision is said to be elastic:

$$a + b \to a + b. \tag{18}$$

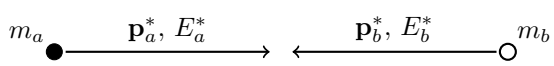
We specify that the excited state of a particle must be considered as a different particle.
We shall consider two reference frames, the CM frame already defined above and the laboratory frame (L). The latter is the frame which, before the collision, one of the particles (called the target) is at rest, while the other (called the beam) moves against it.
Let a be the beam particle, m_a its mass, $\vec{p}_a$ its momentum and E_a its energy; let b be the target mass and m_b its mass.
The following picture shows the system in the initial state



In L, s is given by

$$s = (E_a + m_b)^2 - p_a = m_a^2 + m_b^2 + 2m_b E_a. \tag{19}$$

Let us consider the CM frame, in which the two momenta are equal and opposite.

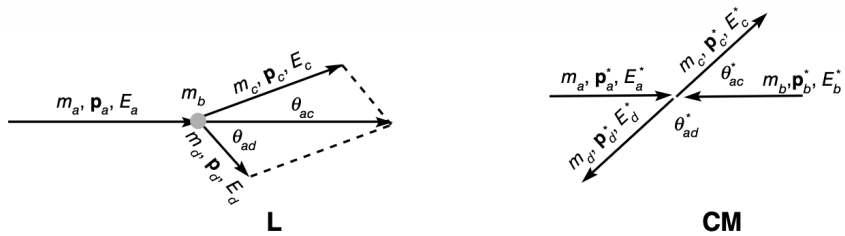


If the energies are larger than the masses, $E_a^* >> m_a$ and $E_b^* >> m_b$, the energies are approximately equal to the momenta: $E_a^* \approx p_a^*$ and $E_b^* \approx p_b^*$. Hence, they are equal to each other, and we call them simply E^*. The total energy squared is

$$s = (E_a^* + E_b^*)^2 \approx (2E^*)^2, \tag{20}$$

where the approximation at the last member is valid for $E^* >> m_a, m_b$.
We see that the total center of mass energy is proportional to the energy of the colliding particles. In the CM frame, all the energy is available for the production of new particles; in the L frame only part of it is available, because momentum must be conserved.
Now let us consider a collision with two particles in final state: this is the two-body scattering

$$a + b \to c + d. \tag{21}$$



this picture shows the initial and final kinematics in the L and CM frames. Notice in particular that, in the CM frame, the final momentum is, in general, different from the initial momentum; they are equal in absolute value only if the scattering is elastic.
Because s is an invariant, it is equal in the two frames; because it is conserved, it is equal in the initial and final states. We have generally in any reference frame

$$s = (E_a + E_b)^2 - (\vec{p_a} + \vec{p_b})^2 = (E_c + E_d)^2 - (\vec{p_c} + \vec{p_d})^2. \tag{22}$$

In two-body scattering, there are two other important kinematic variables that have the dimensions of the square of an energy: the $a - c$ four-momentum transfer t, and the $a - d$ four-momentum transfer u. The first is defined as

$$t = (E_c - E_a)^2 - (\vec{p_c} - \vec{p_a})^2. \tag{23}$$

It is easy to see that the energy and momentum conservation implies that

$$t = (E_c - E_a)^2 - (\vec{p_c} - \vec{p_a})^2 = (E_d - E_b)^2 - (\vec{p_d} - \vec{p_b})^2. \tag{24}$$

In a similar way

$$u = (E_d - E_a)^2 - (\vec{p_d} - \vec{p_a})^2 = (E_c - E_b)^2 - (\vec{p_c} - \vec{p_b})^2. \tag{25}$$

The three variables are not independent. It is easy to show that

$$s + t + u = m_a^2 + m_b^2 + m_c^2 + m_d^2. \tag{26}$$

Notice, finally, that

$$t \leq 0, \qquad u \leq 0. \tag{27}$$

1.4 System of interacting particles

Let us consider a system of interacting particles. We immediately stress that its total energy in not, in general, the sum of the energies of the the single particles, because the field is responsible for the interaction itself contains energy. Similarly, the total momentum is not the sum of the momenta of the particles because the field contains momentum.
Let us consider a particle moving in an external, given field. This means that we can consider the field independent of the motion of the particle.

Let us start with an atomic electron of charge q_e at a distance r from a nucleus of charge Zq_e. The mass of the nucleus is much larger than that of the electron, hence it is not disturbed by the electron motion. The electron moves in a constant potential $\phi = -\frac{1}{4\pi\epsilon_0}\frac{Zq_e}{r}$. The electron energy is

$$E = \sqrt{m_e^2c^4 + p^2c^2} - -\frac{1}{4\pi\epsilon_0}\frac{Zq_e^2}{r} \approx \frac{p^2}{2m_e} - -\frac{1}{4\pi\epsilon_0}\frac{Zq_e^2}{r},$$

where, in the last expression, we have taken into account that the atomic electron speeds are much smaller than c. The final expression is valid in non-relativistic situations, as is the case in an atom, and it is the Newtonian expression of the energy, apart from the irrelevant constant m_ec^2.
The concept of potential is non-relativistic: we can use it if the speeds are smaller in comparison to c or, in other words, if kinetic energies are much smaller than the masses.

1.5 Decays

Consider, for example, the three-body decay $a \to b+c+d$: again, the final state can be defined with more or fewer details, depending on what is measured. Here the quantity to compute is the decay rate in the measured final state. Integrating over all the possible kinematic configurations, one obtains the partial decay rate Γ_{bcd}, or partial with of a in the bcd channel. The sum of all the partial decay rates is the total width of a. The latter is the reciprocal of the lifetime: $\Gamma = 1/\tau$. The branching ratio of a in bcd is the ratio $R_{bcd} = \Gamma_{bcd}/\Gamma$.
Let us start with the collisions, more specifically the "fixed target" collisions. There are two elements:

- The beam: the beam intensity I_b is the number of incident particles per unit time, the beam flux Φ_b is the intensity per unit normal section.
- The target: let n_t be the number of scattering centers per unit volume and let N_t be their total number (if the beam section is small than that of the target, N_t, is the number of centers in the beam section).

The interaction rate R_i is the number of interactions per unit time. By definition of the cross-section σ of the process, we have

$$R_i = \sigma N_t \Phi_b = W N_t, \tag{28}$$

where W is the rate per particle in the target (To be rigorous, one should consider that the incident flux diminishes with increasing penetration depth in the target, due to the interactions of the beam particles).

Consider a beam if initial intensity I_0 entering a long target of density ρ. Let z be the distance traveled by the beam in the beam in the target, measured from its entrance point. We want to to find the beam intensity $I(z)$ as a function of the distance.

Consider a generic infinitesimal layer between z and $z + dz$. If dR_i is the total number of interactions per unit time in the layer, the variation of the intensity in crossing the layer is $dI(z) = -dR_i$. If Σ is the normal section of the target, $\Phi_b(z) = I(z)/\Sigma$ is the flux and σ_{tot} is the total cross-section, we have

$$dI(z) = -dR_i = -\sigma_{tot}\Phi_b(z)dN_t = -\sigma_{tot}\frac{I(z)}{\Sigma}n_t\Sigma dz$$

or

$$\frac{dI(z)}{I(z)} = -\sigma_{tot}n_t dz.$$

In conclusion, we have

$$I(z) = I_0 e^{-n_t\sigma_{tot}z}. \tag{29}$$

The "absorption length", defined as the distance at which the beam intensity is reduced by the factor $1/e$, is

$$\mathcal{L}_{abs} = \frac{1}{n_t\sigma_{tot}}. \tag{30}$$

Another related quantity is the "luminosity", often given in $[cm^{-2}s^{-1}]$, defined as the number of collisions per unit time and unit cross-section:

$$\mathcal{L} = \frac{R_i}{\sigma}. \tag{31}$$

Let I be the number of incident particles per unit time and Σ the beam section; then $I = \Phi_b\Sigma$. $R_i = \sigma N_t\Phi_b = WN_t$ gives

$$\mathcal{L} = \frac{R_i}{\sigma} = \Phi_b N_t = \frac{IN_t}{\Sigma} = In_t l = I\rho N_A 10^3 l, \tag{32}$$

where ρ is the target density.

The Fermi "golden rule" provides the interaction rate W per target particle

$$W = 2\pi|\mathcal{M}_{fi}|^2\rho(E) \tag{33}$$

where E is the total energy and $\rho(E)$ is the phase-space volume available in the final state.
There are two possible expressions of phase-space: the "non-relativistic" expression used is atomic and nuclear physics, and the "relativistic" one used in subnuclear physics.
We recall that in teh non-relativistic formalism the probability that the particle i has the position $\vec{r}_i$ is given by the square modulus of its wave function, $|\psi(\vec{r}_i)|^2$. This is normalized by putting its integral over the entire volume equal to one.
The volume element dV is a scalar in three dimensions, but not is space-time. Under a Lorentz transformation $\vec{r} \to \vec{r}_i$, the volume element changes as $dV \to dV' = \gamma dV$. Therefore, the probability density $|\psi(\vec{r}_i)|^2$ transforms as $|\psi(\vec{r}_i)|^2 \to |\psi(\vec{r'}_i)|^2 = |\psi(\vec{r}_i)|^2/\gamma$. To have a Lorentz-invariant probability

density, we profit from the energy transformation $E \to E^{'} = \gamma E$ and define the probability density as $|(2E)^{-1/2}\psi(\vec{r}_i)|^2$.
The number of phase-space states per unit volume is d^3p_i/h^3 for each particle i in the final state. With n particles in the final state, the volume of the phase-space is therefore

$$\rho_n(E) = (2\pi)^4 \int \prod_{i=1}^{n} \frac{d^3p_i}{(h)^3 2E_i} \delta\left(\sum_{i=1}^{n} E_i - E\right) \delta^3\left(\sum_{i=1}^{n} p_i - \vec{P}\right). \quad (34)$$

Now we consider the collision of two particles, say a and b, resulting in a final state with n particles. We shall give the expression for the cross-section.
The cross-section is normalized to one incident particle; therefore, we must divide by the incident flux. In the laboratory frame, the target particles b are at rest, the beam particles a move with a speed of $\vec{\beta}_a$.
Let us consider, more generally, a frame in which particles b also move, with velocity $\vec{\beta}_b$, which we shall assume parallel to $\vec{\beta}_a$. The flux of particles b is their number inside a cylinder of height $|\vec{\beta}_a - \vec{\beta}_b|$.
If E_a and E_b are the initial energies, the normalization factors of the initial particles are $1/(2E_a)$ and $1/(2E_b)$. It is easy to show that the cross-section is

$$\sigma = \frac{1}{2E_aE_b|\vec{\beta}_a - \vec{\beta}_b|} \int |\mathcal{M}_{\{\rangle}|^2 (2\pi)^4 \prod_{i=1}^{n} \frac{d^3p_i}{(2\pi)^3 2E_i} \delta\left(\sum_{i=1}^{n} E_i - E\right) \delta^3\left(\sum_{i=1}^{n} p_i - \vec{P}\right). \quad (35)$$

The case of a decay is simpler, because in the initial state there is only one particle of energy E. The probability of transition per unit time to the final state f of n particles is

$$\Gamma_{if} = \frac{1}{2E} \int |\mathcal{M}_{\{\rangle}|^2 (2\pi)^4 \prod_{i=1}^{n} \frac{d^3p_i}{(2\pi)^3 2E_i} \delta\left(\sum_{i=1}^{n} E_i - E\right) \delta^3\left(\sum_{i=1}^{n} p_i - \vec{P}\right). \quad (36)$$

The standard model gives the rules to evaluate all the matrix elements in terms of a set of constants.
Now let consider an important case, the two-body phase-space.

Two-body phase-space Let c and d be the two final-state particles of a collision or a decay. We choose the CM frame, in which calculations are easiest. The initial momentum is zero and the initial energy E is known. Then the momenta of the final particles are equal and opposite, $\vec{p}_c = -\vec{p}_d$.